



KUALA LUMPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

(formerly known as Infrastructure University Kuala Lumpur, IUKL)

CSC 2201 – Human Computer Interaction Final Assessment (30%)

EcoPulse

Environmental Awareness Campaign Platform

UI/UX Design Report

Student Name	Mohamed Mohamed Hussein
Student ID	251925403
Exam ID	0315N
Course Code	CSC 2201 – Human Computer Interaction
Submission Date	26 June 2026
Lecturer	Assoc. Prof. Ts. Dr. Robiatul A'dawiah Jamaluddin

Table of Contents

1. Introduction	3
Figma Design Links	3
2. User Experience Planning.....	4
2.1 Project Background and Objective.....	4
2.2 Target Users and User Characteristics	4
2.3 User Goals and Expectations	5
2.4 Human Factors Affecting Interaction.....	6
2.5 Common Usability Issues in Similar Platforms	6
2.6 Design Goals and Usability Objectives	6
3.1 Design Tool Used	9
3.2 Colour Palette and Justification	9
3.3 Typography	10
3.4 Low-Fidelity Wireframes	10
Screen 1 – Home Page (Desktop)	11
Screen 2 – Campaigns List Page (Desktop).....	12
Screen 4 – Join Form (Desktop)	13
Screen 5 – Confirmation Screen (Desktop)	13
Screen 6 – Profile Page (Desktop).....	14
.....	15
Screen 1 – Mobile Splash / Landing Screen.....	15
.....	15
Screen 2 – Mobile Home Dashboard	15
Screen 3 – Mobile Campaigns List.....	16
.....	16
.....	16
Screen 4 – Mobile Campaign Detail	17
.....	17
Screen 5 – Mobile Join Form	17
Screen 6 – Mobile Confirmation Screen.....	18
.....	18
Screen 7 – Mobile Profile Page.....	18
3.7 Navigation Flow	19
3.8 Design Justification.....	19
4. Usability Evaluation and Improvement	20
4.1 Heuristic Evaluation.....	20
4.2 Cognitive Walkthrough.....	21
4.3 Accessibility and Improvement Analysis	22

A. Accessibility Considerations	22
B. Interface Weaknesses Identified	22
C. Before vs Improved Comparison	22
5. Professional Reflection	24
5.1 Design Challenges Encountered	24
5.2 HCI Principles That Influenced My Design.....	24
5.3 Difficulties in Producing a User-Friendly Interaction.....	24
5.4 Most Effective Interface Elements.....	24
5.5 Current Prototype Limitations	25
5.6 Future Improvement Opportunities	25
Appendix A – Figma Prototype Links.....	26
Appendix B – AI Usage Declaration.....	26
Prompt given to Gemini	27
.....	27
AI Response Summary: Gemini provided suggestions for report section ideas, helped check grammar and sentence structure, and suggested ideas for campaign names and platform features. All suggestions were reviewed and modified by the student before being included in the final report.....	27
AI Declaration: The Figma prototype, all screen designs, colour choices, layout decisions, and visual elements were created entirely by me. Gemini was only used as a brainstorming and grammar support tool, not for producing the final written content.....	27
Appendix C – All Final Prototype Screens (Combined View)	27

1. Introduction

Many people want to help the environment, however, they don't know where to begin. People search for campaigns or activities online. However, most of these campaign sites are difficult to navigate and are unenjoyable. This is creating a major obstacle that prevents people from engaging, because although people want to engage with the environment through campaigns, poor design is preventing them. EcoPulse is an answer to this issue. It is a digital platform developed to allow individuals to find environmental campaign opportunities within their own communities. Users will be able to view local campaign opportunities (i.e., tree planting, plastic clean-up, recycling drives) and simply click to join. The platform will track users' individual contributions toward a cleaner environment and reward users with points and badges as motivation.

The document below will outline the entire User Experience / User Interface (UI/UX) design process of EcoPulse. This includes user research, personas, journey mapping, prototype design, usability testing, and professional reflection. The purpose of this document is to demonstrate how well-designed digital engagement platforms can be made more easily accessible, fun, and successful for average consumers.

The design was created using Figma, a professional UI/UX design tool. Two versions of the platform were designed:

- Desktop/Web interface – for users accessing on a laptop or computer
- Mobile interface – for users accessing on a smartphone

Figma Design Links

You can view the original Figma prototypes using the links below:

Desktop Prototype (Figma):

<https://www.figma.com/design/cPHdrHMQPptAuH0JZt9TCt/EcoPulse-Laptop?node-id=0-1&p=f&t=0o5wFq99fDKsDPz2-0>

Mobile Prototype (Figma):

<https://www.figma.com/design/yCWtA8N46FW2VzonCw5sx5/EcoPulse-Mobile?node-id=0-1&p=f&t=IEe8ENQcO0aHmLL3-0>

2. User Experience Planning

2.1 Project Background and Objective

EcoPulse is an environmental awareness campaign platform. The platform is built for a public-facing digital campaign that encourages community members to join and participate in eco-friendly activities such as recycling, tree planting, and beach clean-up events.

Most of today's environmental websites or apps do a poor job of engaging their user base. Environmental sites are cluttered with an overwhelming amount of information and have poorly organized navigation which leaves many feeling confused. EcoPulse was developed to address those issues and make participating easy, rewarding and fun.

Main objectives of EcoPulse:

- Help users easily find and join environmental campaigns near them
- Track and display each user's personal environmental impact
- Motivate users using gamification elements like points and badges
- Make the platform accessible on both desktop and mobile devices
- Provide a clean, simple, and consistent user experience

2.2 Target Users and User Characteristics

EcoPulse is designed for a wide range of users who care about the environment or who want to start making a difference. The main target group includes:

User Group	Description
Young Adults (18–30)	University students and young professionals who are environmentally aware and comfortable using apps and websites.
Community Members (30–50)	Adults who are active in local community events and want to contribute to environmental causes.
Volunteers (All Ages)	People who are already volunteers and want a central platform to manage their activities and track their impact.

2.3 User Goals and Expectations

Based on the target users described above, the following goals were identified:

- Find and explore available environmental campaigns quickly
- Register and join a campaign without confusion
- See their own impact (e.g., trees saved, kg recycled)
- Receive confirmation and feel rewarded after joining
- Use the platform easily on both phone and computer
- Navigate between pages without getting lost

2.4 Human Factors Affecting Interaction

Factors related to humans, i.e., characteristics of people (abilities, behaviors or habits, physical and mental limitations) — that influence how a person will interact with and utilize a product. During the design process for EcoPulse, the following human-related factors were taken into consideration:

Human Factor	How It Was Considered in EcoPulse
Attention	Users have short attention spans. EcoPulse uses large headlines, clear images, and short text to keep the user focused.
Memory	Users should not need to remember information. Campaign details are always visible on the same screen without needing to go back.
Motor Skills	Buttons and tap targets are large enough to be pressed easily, especially on mobile screens.
Perception	Green colour is used throughout to connect the platform visually to nature and the environment. High contrast text makes content easy to read.
Learning	The platform uses familiar patterns (navigation bar, progress bars, profile page) so new users learn quickly without training.

2.5 Common Usability Issues in Similar Platforms

Many environmental and campaign platforms have similar usability problems. These were studied to make sure EcoPulse avoids the same mistakes:

- Too much text on the home page – users get overwhelmed and leave
- Poor navigation – users get lost and cannot find their way back
- No feedback – users do not know if their action (like joining) was successful
- No mobile version – many platforms do not work well on small screens
- Low motivation – platforms do not reward users for participating
- Slow loading or complex registration forms – users give up before completing

EcoPulse was designed to fix all of these issues. The platform has a clean home page, simple navigation, clear confirmation messages, a responsive mobile version, and a gamification system with points and badges.

2.6 Design Goals and Usability Objectives

Design Goal	Usability Objective
Clean and simple home page	User finds what they need in under 10 seconds
Easy campaign joining flow	User completes the join process in 3 steps or fewer

Personal impact tracking	The user can immediately see their contribution
Gamification (points, badges)	User feels motivated to return and join more campaigns
Responsive design	The platform works perfectly on desktop and mobile
Accessibility settings	Users can toggle dark mode, language, and accessibility

2.7 User Persona

A user persona is a fictional but realistic character that represents a typical user of the platform. It helps designers understand who they are designing for and what the user needs, wants, and feels.

PERSONA – Ahmad Farhan	<i>"I want to help the environment, but I need something easy to use."</i>
Age: 24 years old Occupation: University student Location: Kuala Lumpur, Malaysia Tech level: High – uses phone daily	Goals: - Join local eco campaigns - Track his personal impact - Earn recognition for contributing
Frustrations: - Hard to navigate websites - Long, confusing sign-up forms - No feedback after joining	Devices: - Smartphone (primary) - Laptop (secondary) Personality: - Caring, active, social

2.8 User Journey Map

A User Journey Map displays the actions users take to use the product or service in addition to how they think and feel while doing so. The purpose of this is that it enables designers to identify potential issues with the current process and design an improved experience for the user. Below is a representation of Ahmad's journey from discovering EcoPulse and joining his first campaign.

Stage	1 – Awareness	2 – Explore	3 – Decision	4 – Join	5 – Celebrate
Action	Sees EcoPulse post on social media	Opens platform, browses campaigns	Reads Plastic Free Challenge details	Fills form and confirms participation	Sees confirmation + earns 10 points
Feeling	Curious	Interested	Motivated	Confident	Happy and proud

Pain Point	Not sure if platform is trustworthy	Too many choices can be confusing	Wants to know if campaign is near him	Worried if form is too long	Wants to share achievement
Solution	Clean, trustworthy home page with stats	Filter by category + search function	Location + dates shown clearly	Short, simple form – 5 fields only	Success screen + badge animation

2.9 Task Flow Diagram

The task flow diagram shows the exact steps a user takes to complete the main task: joining an environmental campaign. This is the most important user flow in EcoPulse.

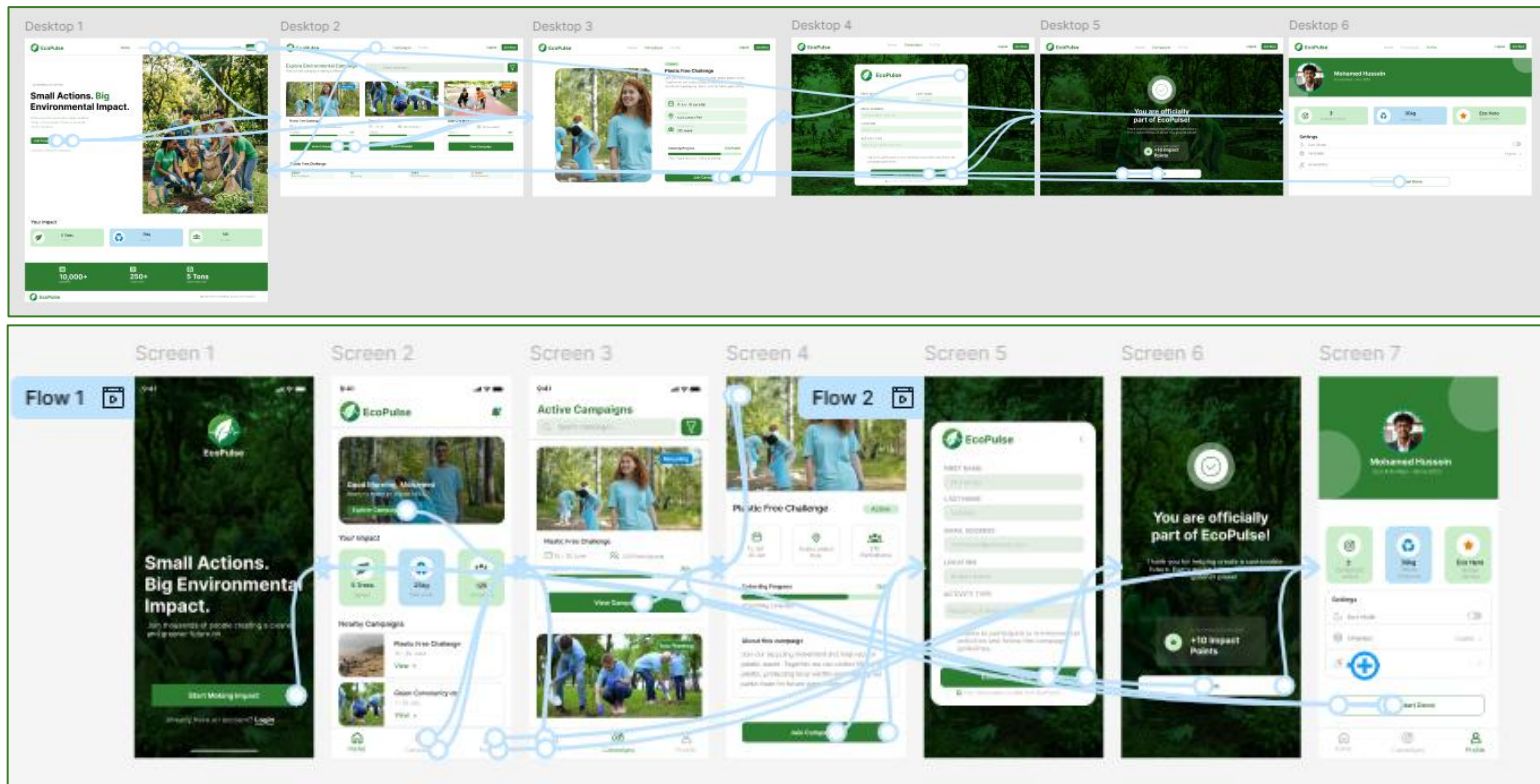
Primary User Task: Join a Campaign

Step	Screen	User Action
1	Home Page	User opens EcoPulse. Sees hero section and statistics. Clicks 'Explore' or 'Join Campaign'.
2	Campaigns List	User sees all active campaigns. Can search by keyword or filter by category (Recycling, Tree Planting, Cleanup).
3	Campaign Detail	User clicks 'View Campaign'. Sees full details: date, location, number of participants, and collecting progress.
4	Join Form	User clicks 'Join Campaign'. Fills in a short form: First Name, Last Name, Email, Location, Activity Type. Tick the agreement checkbox.
5	Confirmation	User clicks 'Confirm Participation'. System shows success screen: 'You are officially part of EcoPulse!' and awards +10 impact points.
6	Profile Page	User sees their updated profile showing campaigns joined, waste reduced, and badges earned.

2.10 Information Architecture / Sitemap

The sitemap shows the structure of the EcoPulse platform — how the pages are organised and connected. A clear information architecture helps users navigate without confusion.

HOME	CAMPAIGNS	PROFILE	JOIN FLOW
• Hero / Tagline • Impact Stats • Community Numbers • CTA Buttons	• Search Bar • Filter Tabs • Campaign Cards • Campaign Detail Page	• User Stats • Badges Earned • Settings • Dark Mode / Language	• Join Form • Participation Agreement • Success / Confirmation • Points Awarded



3. UI/UX Prototype Development

3.1 Design Tool Used

The EcoPulse prototype was designed using Figma. Figma is a cloud-based UI/UX design tool that allows designers to create wireframes, interface screens, and interactive prototypes. It was chosen because:

- It is free and easy to use for students
- It allows designing both desktop and mobile screens in one file
- It supports creating interactive prototypes with clickable links between screens
- It allows easy export of screens as images for reports

3.2 Colour Palette and Justification

The colours used in EcoPulse were carefully chosen to match the platform's theme and purpose. Each colour was selected for a specific reason.

Colour	Hex Code	Why This Colour Was Used
Dark Green (Primary)	#1B6B35	Represents nature, environment, and growth. It is EcoPulse's brand colour and is used for buttons, icons, and headings.
White (Background)	#FFFFFF	Creates a clean, modern, and easy-to-read background. White space prevents the screen from feeling crowded.

Light Green (Cards)	#D6EDD9	Used for cards, chips, and soft highlights to maintain the green theme without being too intense.
Dark Green (Header/Text)	#1B4332	Used for important text, table headers, and deep contrast areas to maintain readability and professionalism.
Nature Background	Photo	Real photos of forests, clean parks, and volunteers are used as backgrounds to connect users emotionally to the cause.

The use of green throughout the platform is intentional. Research in psychology shows that green is associated with nature, safety, health, and positive feelings. This makes it the ideal choice for an environmental platform where the goal is to inspire action and create trust.

3.3 Typography

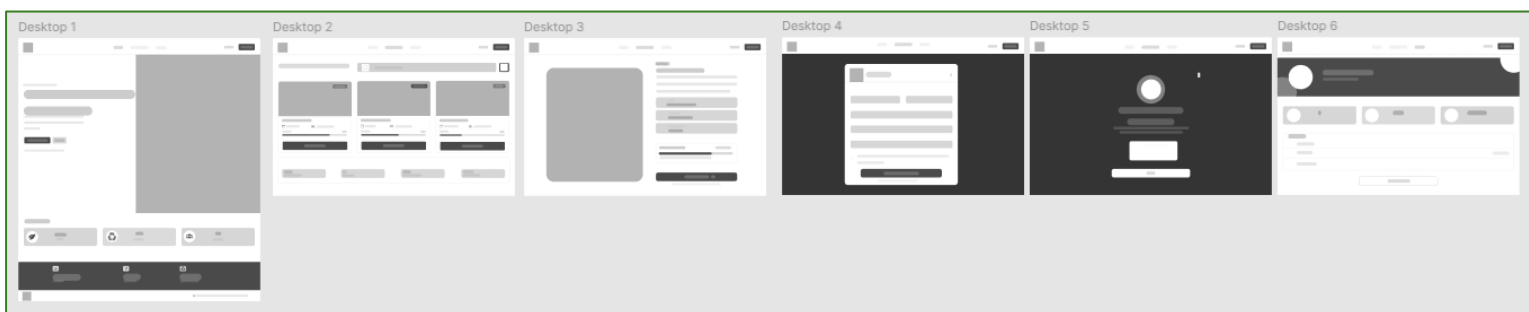
Fonts Used In EcoPulse Were Chosen For Clarity And Readability On All Screen Sizes

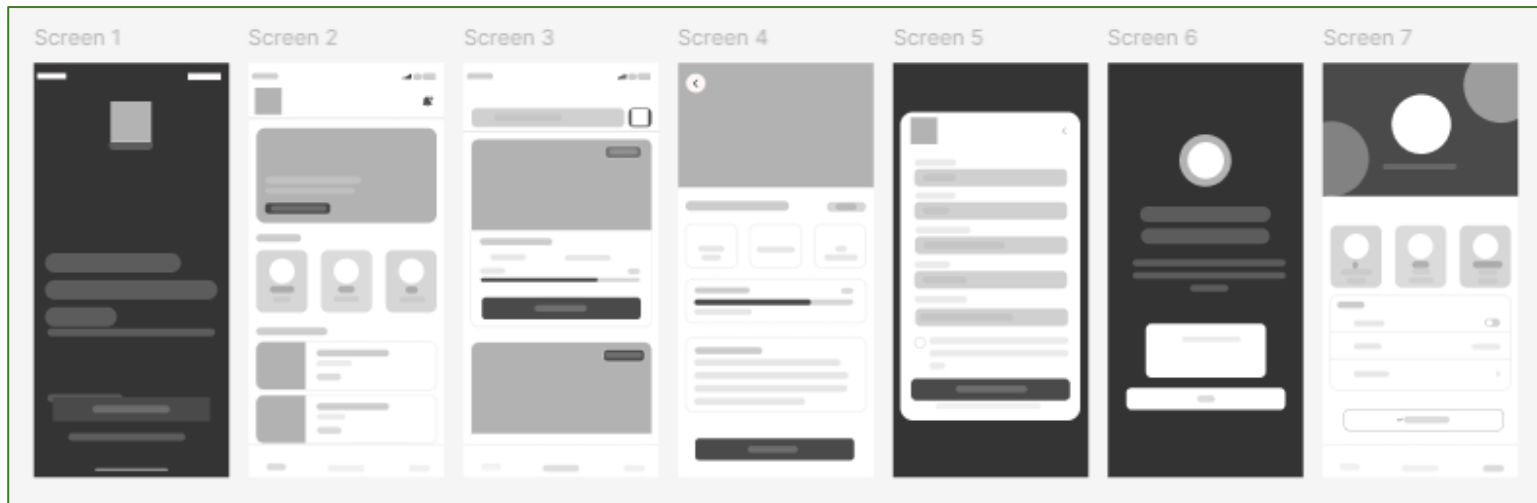
- **Headlines:** Fonts such as "Small Actions. Big Environmental Impact." use very large and bold type to draw your immediate attention. The larger headline font works very well with mobile phones.
- **Body Copy:** An easy-to-read, non-serif font is used for all body copy or description of what you're doing. Body copy is written so it's easily readable at most viewing sizes (14-16px on a computer, 14px on a mobile).
- **Statistics/Number Data:** Bold typeface was utilized (i.e., "10,000+ Members", "5 Tons Waste Reduced") to create an instant visual representation of impact. There is research that indicates when people see visible statistics, they react in a positive manner.

3.4 Low-Fidelity Wireframes

Before designing the final screens, low-fidelity wireframes were developed. Low-fidelity wireframes are basic sketches that display the arrangement and organization of all elements on a page with no color, images or details of design. This type of sketch is used to determine where each element will be placed.

The wireframes below show the early layout planning for the main screens:



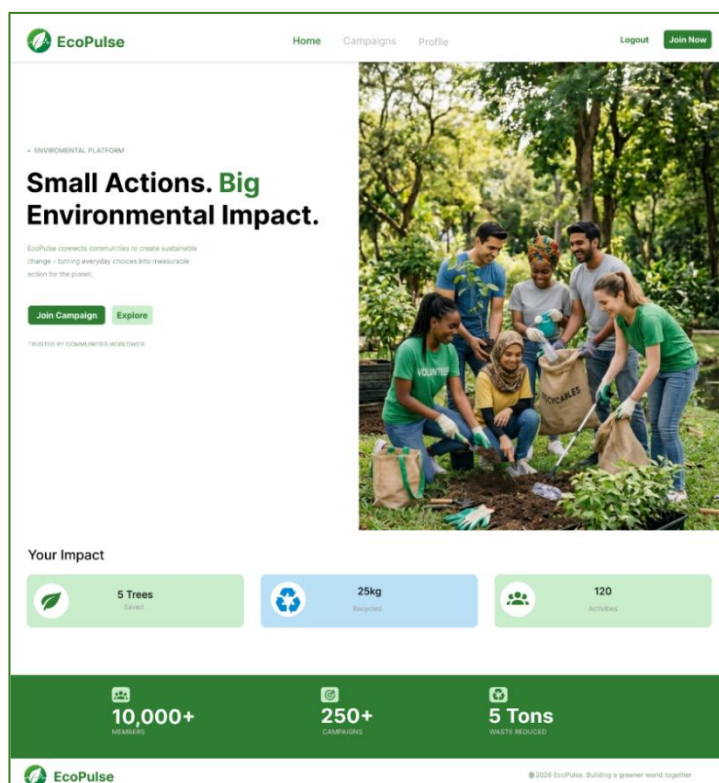


3.5 Final Interface Screens – Desktop Version

The following screenshots show the completed desktop interface of EcoPulse. All screens maintain a consistent layout, colour scheme, and navigation bar.

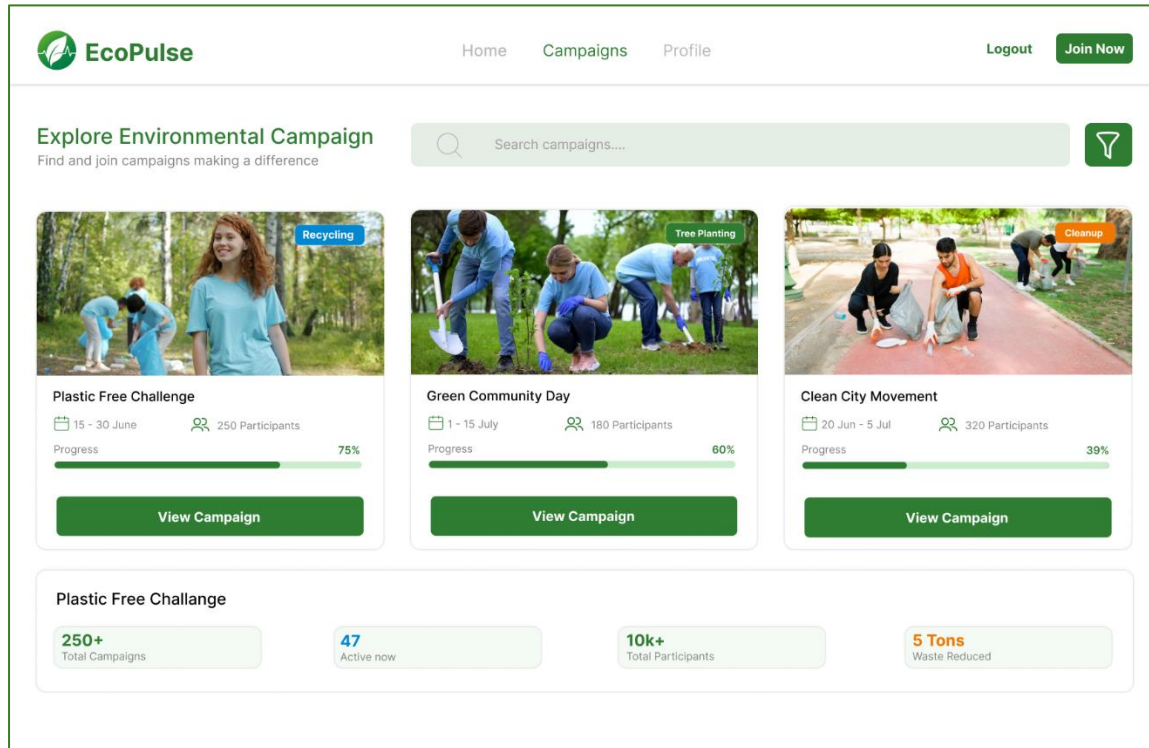
Screen 1 – Home Page (Desktop)

The main website uses a compelling tag line for EcoPulse: "Little actions can have a major environmental impact." This is accompanied by a strong emotional appeal through the use of a high-quality, full-screen natural-environment photo. Beneath this Hero Section, your personal impact on the environment is shown in numbers (e.g., trees you've saved, kilograms you've recycled, etc.). At the bottom, global community statistics are showing the total number of members and active campaigns to establish credibility.



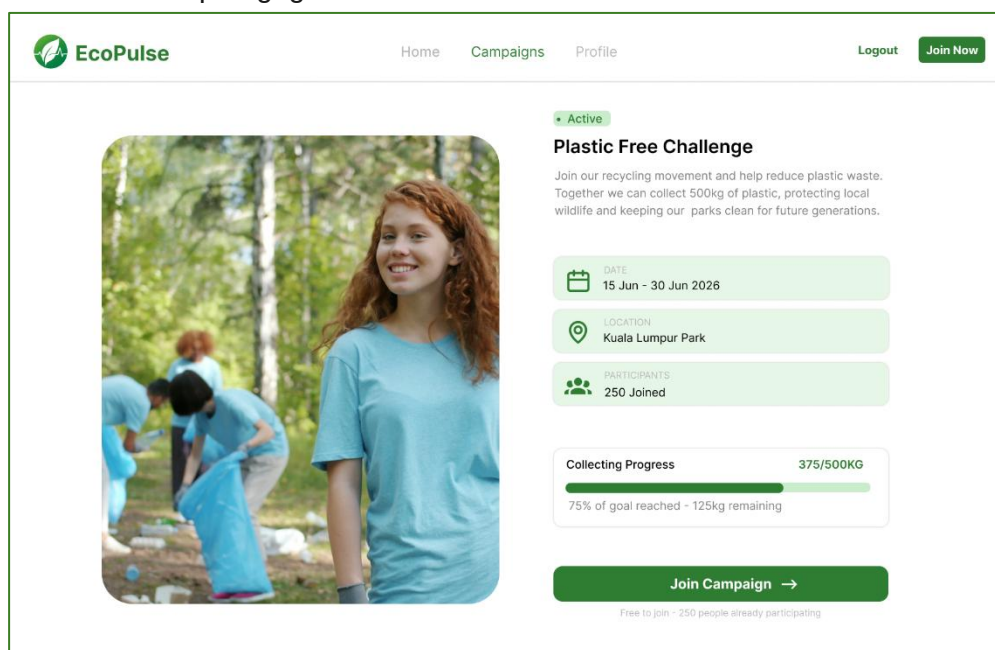
Screen 2 – Campaigns List Page (Desktop)

Beginning with the campaigns page, users see a list of environmental campaigns that are currently active in the form of cards. On this screen, there is a "search" area to find specific campaigns based on what you want to find, as well as tabs above it (Progress, Recycling, Tree Planting, Cleanup), which allow you to sort the campaigns by category.



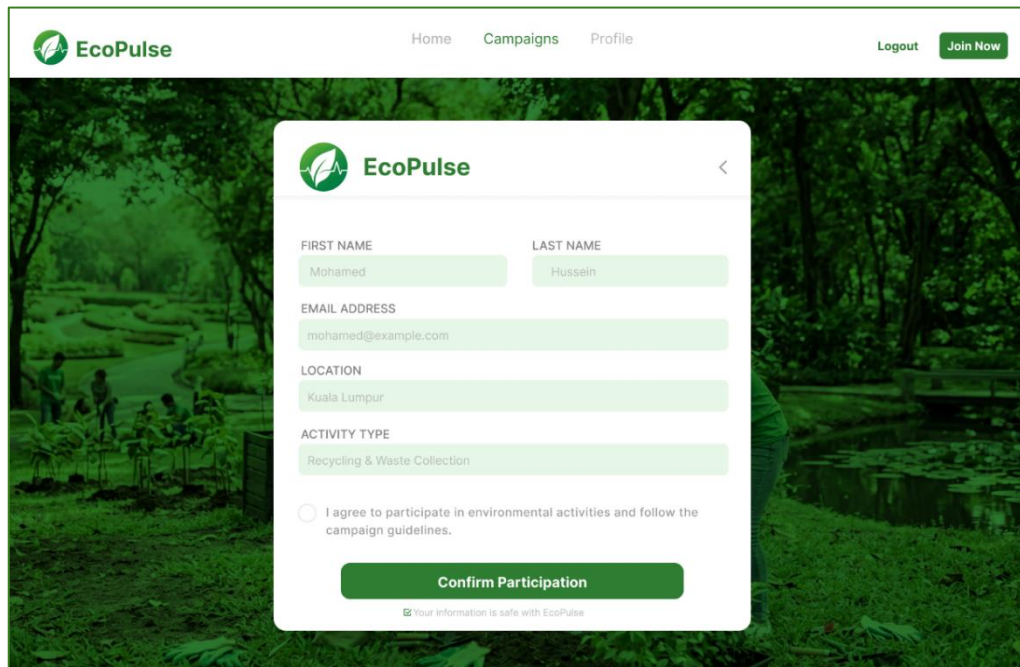
Screen 3 – Campaign Detail Page (Desktop)

When users click "View Campaign," the full campaign information is displayed for them. The campaign information will include campaign name & status, start/end date, where it is happening, how many people are participating in the campaign, a visual representation of collection progress (eg, 375/500kg collected – 75%), the campaign description & a "Join Campaign" link. On the right-hand side of this page, there is a picture of a real nature setting, which can help engage users.



Screen 4 – Join Form (Desktop)

The join form is brief. It will have First Name, Last Name, Email Address, Location, and Activity Type boxes. At the end of the form, there will be a check box for the user to indicate that they have read the campaign guidelines. The "Confirm Participation" button will be prominent on the page. In addition, as an additional comfort to the user, it will say "Your Information is Safe with EcoPulse" at the bottom of the form.

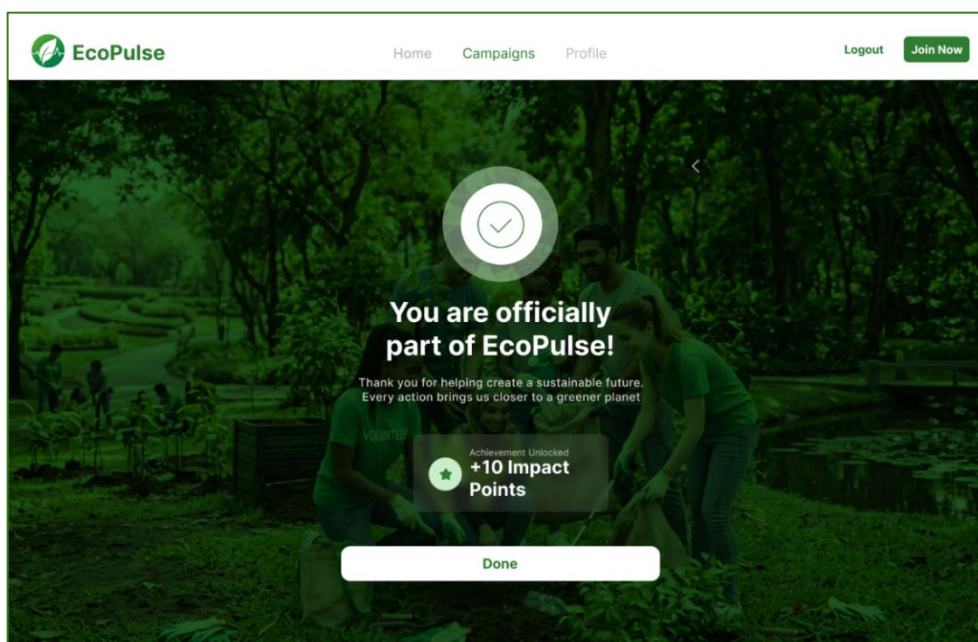


The screenshot shows the EcoPulse website's join form. The form is centered on a green background image of people planting trees. The form fields are as follows:

- FIRST NAME:** Mohamed
- LAST NAME:** Hussein
- EMAIL ADDRESS:** mohamed@example.com
- LOCATION:** Kuala Lumpur
- ACTIVITY TYPE:** Recycling & Waste Collection
- ☐ I agree to participate in environmental activities and follow the campaign guidelines.
- Confirm Participation** (green button)
- ✔ Your information is safe with EcoPulse

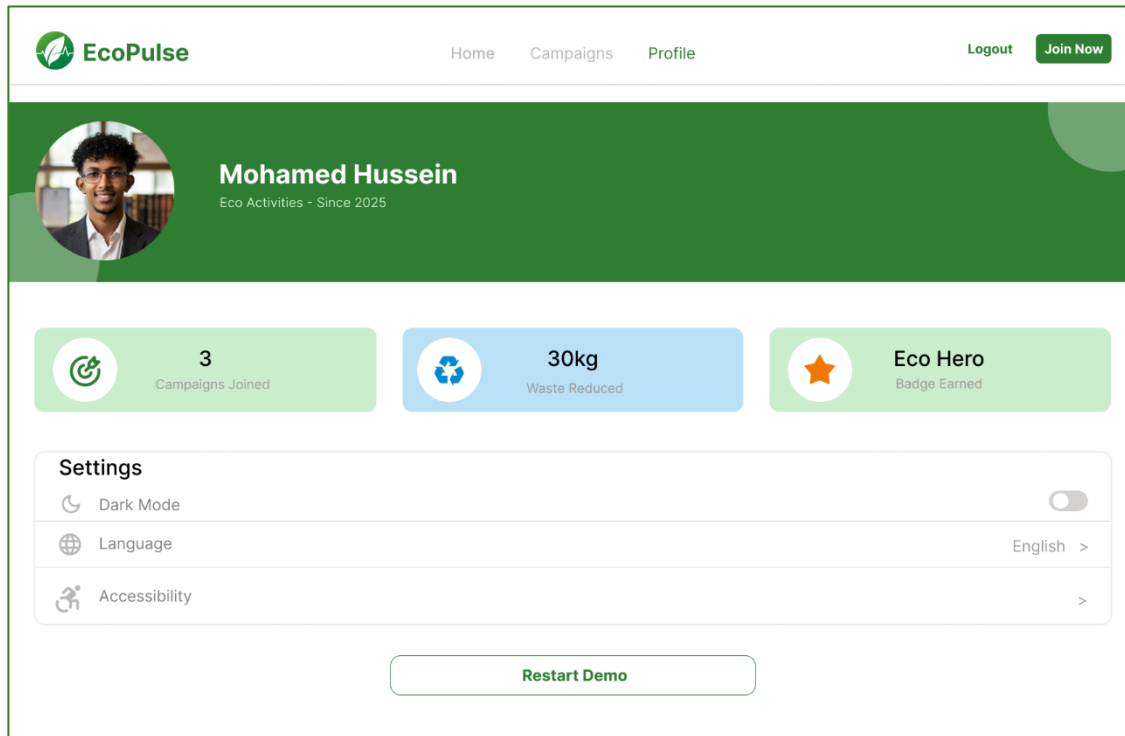
Screen 5 – Confirmation Screen (Desktop)

Once you've completed the sign-up form, users are redirected to an affirmation screen which reads "you're now officially a part of EcoPulse!" along with a fun green check mark animation. Users earn +10 Impact points in a rewards box from your campaign. Users will receive instant positive reinforcement and be encouraged to return to other future campaigns.



Screen 6 – Profile Page (Desktop)

The profile allows the user to see their own personal stats: campaigns joined, waste reduced, and badges earned (Eco hero), in addition to an option to customize the website with Dark mode, language preference, and accessibility options, as seen below. It adds a personal touch to the app, which encourages people to continue using the app.



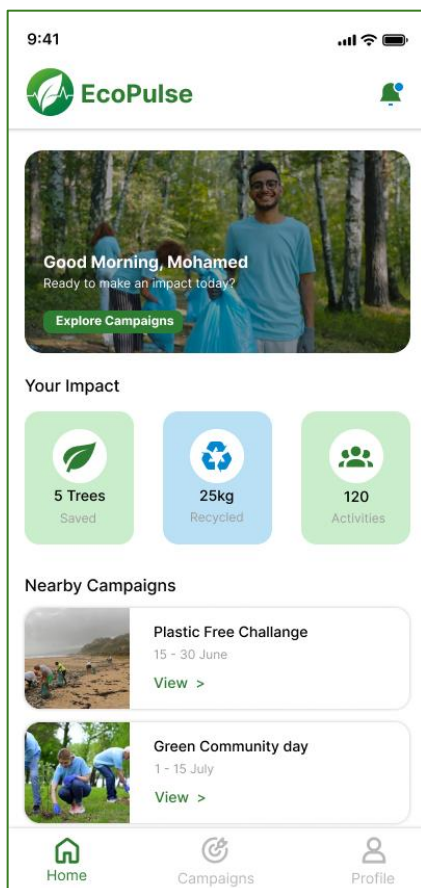
3.6 Final Interface Screens – Mobile Version

The mobile version of EcoPulse adapts the same design to fit a smartphone screen. The layout changes to a vertical scroll format with a bottom navigation bar (Home, Campaigns, Profile) replacing the top navbar.



Screen 1 – Mobile Splash / Landing Screen

The mobile app opens with a full-screen nature background with the EcoPulse logo and tagline. A large green 'Start Making Impact' button calls the user to action. The layout is clean and simple, perfect for a small screen.

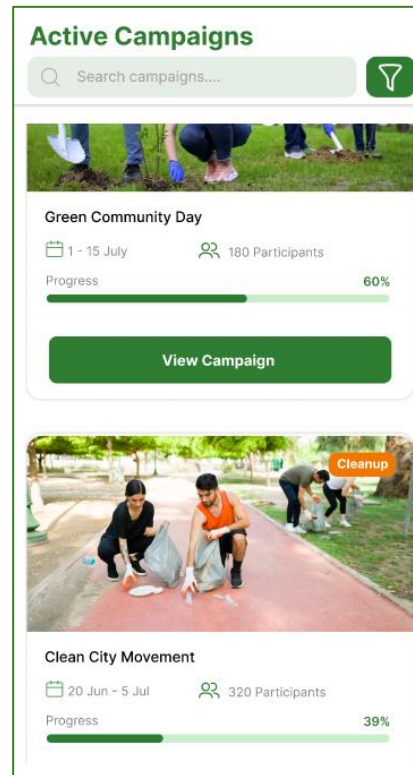
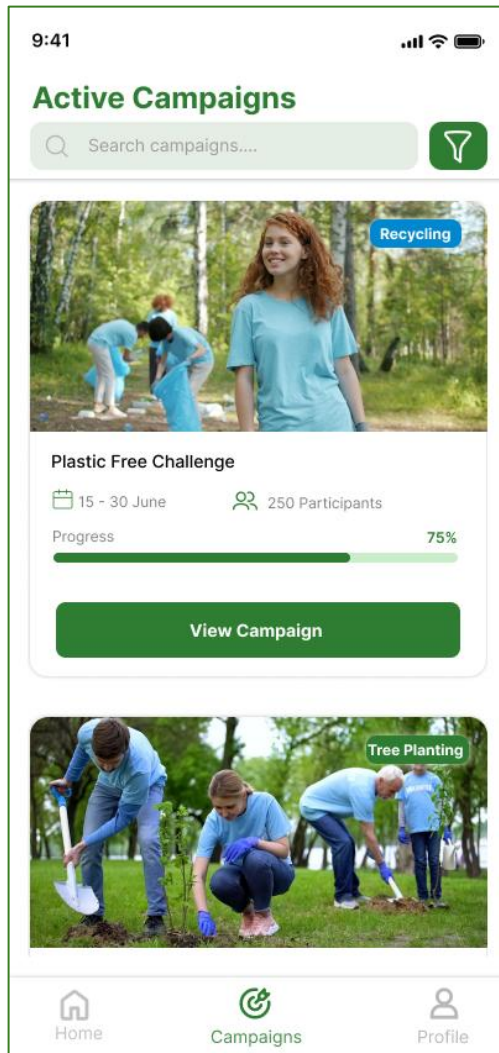


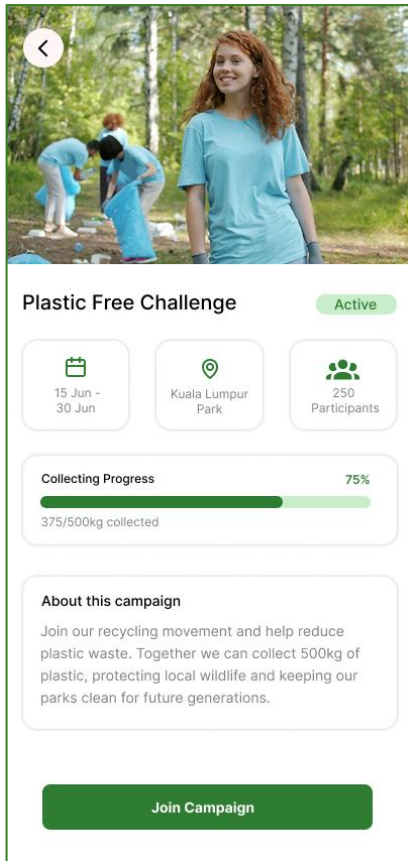
Screen 2 – Mobile Home Dashboard

After logging in, the user sees a personalized dashboard with a greeting ('Good Morning, Mohamed'), an 'Explore Campaigns' card, personal impact stats, and a list of nearby campaigns. The bottom navigation bar is always visible for easy access.

Screen 3 – Mobile Campaigns List

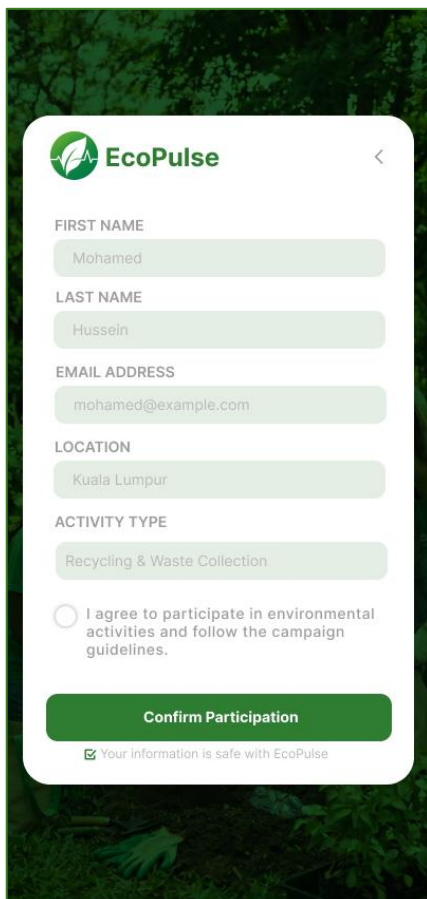
The campaigns list on mobile uses the same category filters and progress bars as the desktop version, but adapted to a scrollable vertical list. Each campaign card is full-width and easy to tap.





Screen 4 – Mobile Campaign Detail

The campaign detail screen on mobile shows all the information in a scrollable format. Key details (date, location, participants) are shown as icon chips at the top. The collecting progress bar and campaign description follow below. A full-width 'Join Campaign' button is at the bottom for easy access.



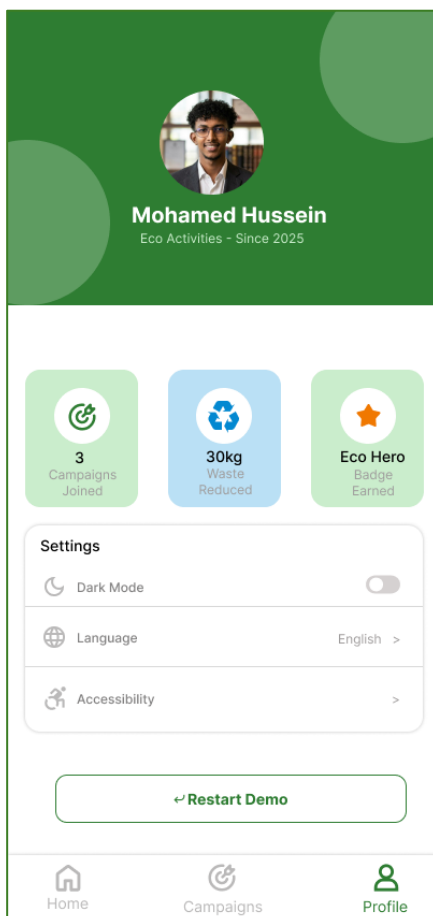
Screen 5 – Mobile Join Form

The mobile join form uses the same simple layout as the desktop but is optimized for touch input. Each field is large enough to tap easily. The 'Confirm Participation' button stretches the full width of the screen.



Screen 6 – Mobile Confirmation Screen

The confirmation screen on mobile shows the same success message and +10 Impact Points award. The full dark green background creates a strong visual reward moment that feels satisfying on mobile.



Screen 7 – Mobile Profile Page

The mobile profile page shows the user's avatar, name, campaign statistics (Campaigns Joined, Waste Reduced, Eco Hero badge), and settings. The settings include a Dark Mode toggle, a Language selector, and an Accessibility toggle.

3.7 Navigation Flow

EcoPulse uses a simple and consistent navigation structure across both desktop and mobile. The user always knows where they are and how to go back.

Desktop Navigation	Location	Mobile Navigation	Location
Home	Top navbar (left logo, right links)	Home	Bottom tab bar icon
Campaigns	Top navbar link	Campaigns	Bottom tab bar icon
Profile	Top navbar link	Profile	Bottom tab bar icon
Join Campaign	Button on campaign detail page	Join Campaign	Button at bottom of detail screen

3.8 Design Justification

Every design decision in EcoPulse was made with the user in mind. The following explains the main design choices and why they were made:

Design Decision	Justification
Green colour scheme	Green psychologically represents nature, trust, and growth. It connects users emotionally to the environmental mission of EcoPulse.
Nature photography	Real photos of forests, volunteers, and clean environments help users feel connected to the cause and build trust in the platform.
Short join form (5 fields)	Long forms cause users to abandon the process. Keeping it to 5 fields removes friction and increases completion rates.
Progress bars	Progress bars show how close a campaign is to its goal. This motivates users to join and contribute before the deadline.
Points and badges (gamification)	Gamification increases user engagement and repeat visits. Earning points and badges makes participation feel rewarding.
Bottom nav bar (mobile)	The bottom navigation is the standard pattern for mobile apps (used in Instagram, WhatsApp, etc.). Users already know how to use it without learning.
Confirmation screen	Giving immediate positive feedback after an action (Nielsen's Heuristic #1: Visibility of System Status) reassures users that their action was successful.

4. Usability Evaluation and Improvement

4.1 Heuristic Evaluation

A heuristic evaluation is a usability inspection method. It involves checking a design against a set of recognized usability rules, called heuristics. The most widely used heuristics were created by Jakob Nielsen in 1994. There are 10 heuristics in total. EcoPulse was evaluated against all 10.

Severity scale: 0 = No problem, 1 = Minor problem, 2 = Moderate problem, 3 = Major problem

#	Heuristic	Observation in EcoPulse	Severity	Status
1	Visibility of System Status	The confirmation screen clearly tells users they have joined. Progress bars show campaign status. Points are awarded immediately.	0	✓ Pass
2	Match Between System and Real World	The language used is simple and friendly. Icons match real-world concepts (leaf for nature, recycle icon). The campaign card design resembles event posters.	0	✓ Pass
3	User Control and Freedom	Users can navigate freely using the navbar. However, there is no 'cancel' or 'back' button visible in the join form. This could be improved.	2	⚠ Improve
4	Consistency and Standards	Green colour, font sizes, button styles, and card layouts are consistent across all screens and both desktop and mobile versions.	0	✓ Pass
5	Error Prevention	The form uses clear field labels and placeholder text to guide users. The agreement checkbox prevents accidental submissions. However, no live validation is shown for empty fields.	1	⚠ Minor
6	Recognition Rather Than Recall	Campaign details (date, location, participants) are always visible on screen. Users do not need to memorise anything from a previous step.	0	✓ Pass
7	Flexibility and Efficiency	The search bar and category filters help experienced users find campaigns quickly. Nearby campaigns are shown on the home dashboard for quick access.	0	✓ Pass

8	Aesthetic and Minimalist Design	Each screen shows only the information that is needed. There is no clutter. White space is used generously. The design feels clean and modern.	0	✓ Pass
9	Help Users Recognise, Diagnose and Recover from Errors	No error messages are currently shown if a form field is left empty. This is a missing feature that should be added in a future version.	2	⚠ Improve
10	Help and Documentation	The platform does not currently have a help section or FAQ. For a simple platform like EcoPulse, most tasks are self-explanatory, but this could be added later.	1	⚠ Minor

4.2 Cognitive Walkthrough

A cognitive walkthrough is a usability evaluation method where we trace through a specific task step by step and ask: Can the user figure out what to do next? After each action, do they get clear feedback? EcoPulse was evaluated on the task: 'Join the Plastic Free Challenge Campaign'.

Step	User Action	Can user figure it out?	Feedback Given?	Issue / Note
1	Opens home page	✓ Yes – tagline is clear	Hero section with CTA buttons	No issues
2	Clicks 'Explore' or navigates to Campaigns	✓ Yes – navbar and CTA buttons are visible	Campaigns page loads with cards	No issues
3	Finds 'Plastic Free Challenge' card	✓ Yes – category filters help narrow results	Campaign cards show dates and progress	User might miss filter tabs if not highlighted
4	Clicks 'View Campaign'	✓ Yes – button is clearly visible and labelled	Detail page opens with full info	No issues
5	Clicks 'Join Campaign'	✓ Yes – button is large and in primary green colour	Join form appears	No issues
6	Fills in form fields	⚠ Mostly yes – some users may not know what 'Activity Type' means	Placeholder text guides input	A tooltip or example text for 'Activity Type' would help

7	Ticks agreement checkbox and clicks 'Confirm'	✓ Yes – checkbox and button are visible	Confirmation screen appears with success message	No back button to cancel – improvement needed
8	Sees '+10 Points' achievement	✓ Yes – achievement is large and clear	Points displayed prominently on confirmation screen	No issues – gamification works well here

Overall, the cognitive walkthrough shows that the Join Campaign flow is clear and easy to complete. Two minor issues were identified: a missing tooltip for the 'Activity Type' field, and the absence of a 'Cancel' or 'Back' button on the join form.

4.3 Accessibility and Improvement Analysis

A. Accessibility Considerations

EcoPulse was designed with accessibility in mind. The following features were included:

- Dark Mode toggle in the profile settings – helps users with light sensitivity
- Language selection option – supports non-English speaking users
- Accessibility settings toggle – reserved for future features like larger text or screen reader support
- High contrast text – all text is dark green or dark grey on white/light backgrounds
- Large tap targets – all buttons and form fields are large enough for users with motor difficulties
- Simple language – all content is written in simple English so users of all backgrounds can understand

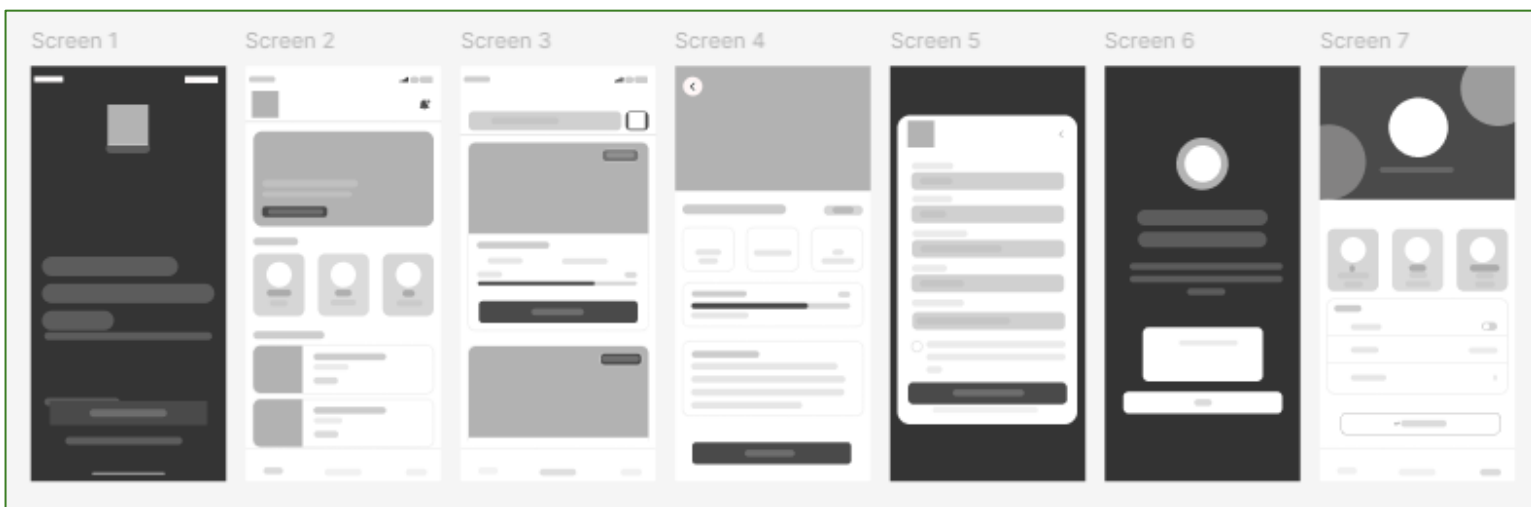
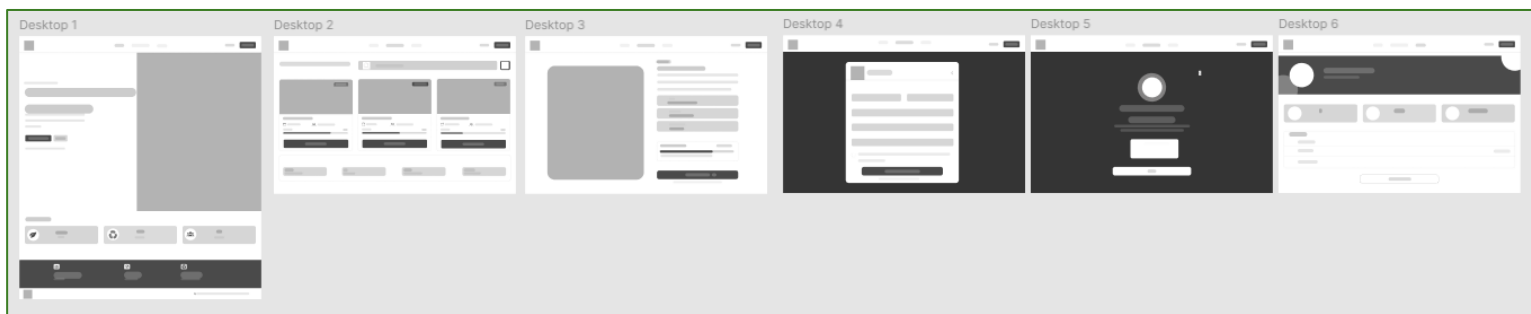
B. Interface Weaknesses Identified

- No form validation messages – if a user leaves a field empty, there is no visible error message
- No cancel or back button on the join form
- No help section or FAQ page
- Spelling errors in the original screens (Environmental, Challenge, Guidelines, Planet, Accessibility) – these should be corrected

C. Before vs Improved Comparison

Area	Before (Original Issue)	After (Improved Version)
Join Form	No cancel button. User cannot go back once they start filling the form.	Add a visible 'Cancel' or 'X' button at the top of the form to give users an easy exit.
Error Handling	If a form field is empty and user clicks 'Confirm', nothing happens or the form submits incorrectly.	Show a red border and an error message below empty required

		fields (e.g. 'This field is required').
Activity Type Field	Label 'Activity Type' is vague. New users do not know what to enter.	Add a dropdown menu with preset options (Recycling, Tree Planting, Clean-up, etc.) with a small tooltip icon.
Spelling	'Enviromental', 'Challange', 'guidliness', 'planed', 'Accesibility' – multiple spelling errors visible in screens.	Corrected to: Environmental, Challenge, Guidelines, Planet, Accessibility.



5. Professional Reflection

5.1 Design Challenges Encountered

Designing EcoPulse was an enjoyable and difficult experience. Perhaps my greatest difficulty came from trying to get the platform to function correctly on two different types of screens: desktop and mobile. Desktop screens are large (usually) with ample width for placing elements side-by-side. Mobile screens are narrow (usually) with limited space to display multiple elements. As such, when designing each type of screen, I asked myself: How does this look on a small mobile phone? This required that I consider the implications of responsive design as early as possible while creating the visual design for EcoPulse. Another significant challenge I experienced was balancing simplicity in the design and avoiding a "blank" feeling. Initially, I placed too many elements onto the homepage — too many sections, too many buttons. Upon review, I removed unnecessary elements and focused only on those items that were most critical to users' understanding of their role within EcoPulse: the tagline, the campaigns they supported, and the impact they created.

5.2 HCI Principles That Influenced My Design

Several HCI principles guided the design decisions in EcoPulse:

- Nielsen's Heuristic 1 (Visibility of System Status): I added progress bars, participant counts, and a confirmation screen so users always know what is happening on the platform.
- User-Centered Design (UCD): I started by thinking about my users — young adults and community volunteers — before designing anything. Their goals (find campaigns, join easily, track impact) shaped every screen.
- Affordance: Buttons are large and clearly clickable. Campaign cards look like real-world event posters, which makes users instinctively know what to do with them.
- Consistency: The same fonts, colours, icons, and navigation structure are used on every screen, so users only need to learn the interface once.

5.3 Difficulties in Producing a User-Friendly Interaction

One problem with this part was the sign-up form. I wanted it as short as possible, but there had to be enough information in the 5 fields that would allow me to properly register the user. It took some thought, but ultimately I decided to go with just 5 fields and replace an extensive terms and conditions page with a simple checkbox. The result was a much cleaner and quicker experience for filling out the form.

Another challenge was designing the confirmation page. I didn't want it to look anything like your typical "Thank you" page. I wanted my users to actually feel excited when they received their confirmation and were rewarded. By adding "+10 Impact Points", which is our gamification component, and the check mark icon, we felt the confirmation looked exciting and motivating.

5.4 Most Effective Interface Elements

Looking at the overall design, the most effective elements in EcoPulse are:

- The progress bars on campaign cards – they create urgency and motivate users to join before a goal is reached

- The personal impact dashboard on the home page – showing '5 Trees Saved, 25kg Recycled, 120 Activities' makes users feel their contributions matter
- The confirmation screen with achievement points – this was the most rewarding moment in the user flow, and it encourages return visits
- The bottom navigation bar on mobile – it follows the standard mobile app pattern and makes navigation completely intuitive

5.5 Current Prototype Limitations

- No real-time form validation – the join form does not show error messages for empty fields
- No cancel button in the join form – users cannot exit the form without using the back button
- No search functionality implemented in the prototype – the search bar is visible but not interactive
- Spelling errors found in the original Figma screens – these were identified during evaluation and listed for correction
- No help page or FAQ – new users have no documentation if they are confused

5.6 Future Improvement Opportunities

- Add live form validation with clear red/green border feedback on each field
- Add a 'Cancel' button on the join form and a confirmation dialog ('Are you sure you want to leave?')
- Develop a fully interactive search and filter function for campaigns
- Add a notification system to remind users about upcoming campaign dates
- Build a social sharing feature so users can share their achievements on social media
- Conduct a user testing session with real users (at least 5 participants) to collect feedback and further improve the design

6. Conclusion

EcoPulse is a user-centered, community-based campaign platform that connects people to meaningful eco-activities. The HCI (Human Computer Interaction) Life-Cycle Design Process was used to develop EcoPulse. This involved user research and planning, then developing prototypes, testing usability, and providing professional reflections.

Seven desktop and seven mobile screens were developed by using Figma as the tool for the UI/UX design. These screens provide the complete user-flow from the landing-page to confirming campaign participation. A consistent green color scheme, clear navigation options, short-forms, and gamification elements are incorporated into the design to create a simple, fun and rewarding experience.

An analysis of EcoPulse against Nielsen's ten usability principles shows that EcoPulse meets all but two of them. Some minor adjustments need to be made to improve error-handling and user-control. The "Join Campaign" workflow was clearly defined during the Cognitive Walkthrough, which demonstrated that this workflow would be easily completed by users.

I learned from working on this project the importance of designing for the user, not just creating something aesthetically pleasing, but making sure it functions well. Good User Experience/UI design does not simply consist of colors or layout; it includes understanding

users' motivations, needs and frustrations, and removing obstacles and building positive experiences.

7. References

Interaction Design Foundation. (2023). User Experience (UX) Design. <https://www.interaction-design.org/literature/topics/ux-design>

Nielsen, J. (1994). 10 usability heuristics for user interface design. Nielsen Norman Group. <https://www.nngroup.com/articles/ten-usability-heuristics/>

Nielsen, J., & Mack, R. L. (Eds.). (1994). Usability inspection methods. John Wiley & Sons.

Norman, D. A. (2013). The design of everyday things (Revised and expanded ed.). Basic Books.

Preece, J., Rogers, Y., & Sharp, H. (2019). Interaction design: Beyond human-computer interaction (5th ed.). John Wiley & Sons.

Figma. (2024). Figma design tool overview. <https://www.figma.com>

Wroblewski, L. (2011). Mobile first. A Book Apart.

World Wide Web Consortium (W3C). (2023). Web content accessibility guidelines (WCAG) 2.1. <https://www.w3.org/TR/WCAG21/>

8. Appendix

Appendix A – Figma Prototype Links

The complete interactive Figma prototypes can be accessed using the following links:

Desktop Prototype: [Desktop Prototype](#)

Mobile Prototype: [Mobile Prototype](#)

Appendix B – AI Usage Declaration

In accordance with the AI Usage Guidelines stated in the assessment brief, AI tools were used in this project ONLY for the following purposes:

- Brainstorming content structure and section ideas
- Grammar and language checking
- Generating initial ideas for campaign names and platform features

All content was reviewed, verified, personalised, and modified by me. No AI-generated content was submitted directly without modification. Screenshots of the AI tool usage are shown below.



Prompt given to Gemini

I have a final university assignment for CSC 2201 Human Computer Interaction. I designed a UI/UX prototype called EcoPulse — an environmental awareness campaign platform — using Figma. I need help structuring my report. The report needs to cover user experience planning, prototype development, usability evaluation, and professional reflection. Can you help me with ideas for section content, grammar checking, and campaign name suggestions?



Pro



AI Response Summary:

Gemini provided suggestions for report section ideas, helped check grammar and sentence structure, and suggested ideas for campaign names and platform features. All suggestions were reviewed and modified by the student before being included in the final report.

AI Declaration:

The Figma prototype, all screen designs, colour choices, layout decisions, and visual elements were created entirely by me. Gemini was only used as a brainstorming and grammar support tool, not for producing the final written content.

Appendix C – All Final Prototype Screens (Combined View)

The following pages show all prototype screens for both desktop and mobile versions of EcoPulse in a combined overview format.

